

Jake Vaughn

Computer Engineer

CONTACT

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Des Moines, IA 

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SKILLS

C#

.Net Maui

TypeScript

Verilog, VHDL

C++

Java

SQL (MySQL, SQLite)

Collegiate Extracurriculars

ISU Mountaineering
and Climbing Club (MCC)

Executive Club Member

Risk Manager

Information Assurance
Student Group (IASG)

Student Group Interested
in Cyber Security

Great Plains Math
League (GPML)
Mathematics Competitions

EDUCATION

Iowa State University, Ames IA
— *B.S. Computer Engineering*

Aug 2017 - May 2021

3.45 GPA

Relevant Coursework: Assembly Level Programming, Software Development Practices, Design/Analysis of algorithms, Cyber Security Concepts, and Embedded Systems Design.

EXPERIENCE

Vermeer — Software Engineer

Jan 2023 - Current

- Building **BorePlan**, a cross-platform mobile application for planning, executing, and documenting horizontal directional drilling (HDD) projects.
- Architected and implemented bore path generation algorithms, enabling precise rod-by-rod HDD planning, and utility collision detection.

Self-Employed — Software Development

July 2020 - Sep 2022

- Developing MEV Bots using Typescript, Node.JS, and Hardhat
- Creating and managing nodes for RPC endpoint connections

PROJECTS

MIPS Processor

Simulation of a processor built from scratch in ModelSim using VHDL on an Altera FPGA board. The processor featured 5 stage pipelining, a custom instruction set, and hazard detection.

Custom ASIC Temperature Sensor

Designed and implemented a custom ASIC to compute the moving average and standard deviation of real-time temperature data using Cadence Genus and Cadence Innovus.

Visualizing Substrate — Network Visualization Tool

TypeScript-based utility that extracts transaction data from Substrate based networks via RPC calls and transforms it into node and edge lists compatible with Gephi for visualization.